ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Constant Acceleration in 1D

One-dimensional motion is a sign convention plus the right SUVAT equation.

Constant Acceleration in 1D · M1 · 04

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 SUVAT: Choose The Missing Letter

TRIGGER *constant acceleration,* u, v, a, s, t

02 SIGN: Sign Convention

TRIGGER *opposite directions, deceleration*

03 MEET: Meeting Motions

TRIGGER *two particles meet, overtake*

04 VERT: Vertical Gravity

TRIGGER *thrown up, falling, maximum height*

BANK EVIDENCE Local pages: 22 problem, 41 answer, 12 notes. Website primary entries: 16. Website linked entries: 66.

01 SUVAT: Choose The Missing Letter

TRIGGER *constant acceleration, u, v, a, s, t*

BECOMES List knowns and choose the equation missing the unwanted letter.

FIRST LINE TO WRITE

$$u,\ v,\ a,\ s,\ t$$

SIMPLEST STRATEGY

1. **S**elect positive direction.
2. **U**nderline known quantities.
3. **V**acant letter is avoided.
4. **A**pply the matching equation.
5. **T**idy units.

WORKED MODEL

> If u, a, t are known and s is needed, use $s = ut + \frac{1}{2}at^2$.

HARD VARIANTS

1. Only use SUVAT when acceleration is constant.
2. If direction changes, split the motion at $v = 0$.
3. Convert units before writing the equation.

BOTTOM LINE

SUVAT is a selection problem first.

Practice Page: SUVAT

QUESTION TYPE SUVAT

A particle starts from rest and accelerates at 3 $\mathrm{m\,s^{-2}}$ for 5 s.

COMMON TRAP Choosing an equation before listing knowns.

02 SIGN: Sign Convention

TRIGGER *opposite directions, deceleration*

BECOMES Choose one positive direction and make all quantities obey it.

FIRST LINE TO WRITE

> up/right/forward is positive

SIMPLEST STRATEGY

1. **S**et the positive direction.
2. **I**nsert signs for u, v, a, s.
3. **G**uard against word signs.
4. **N**ote final direction.

WORKED MODEL

> If upward is positive, gravity gives $a = -g$.

HARD VARIANTS

1. Displacement can be negative even when distance is positive.
2. Final velocity sign tells direction.
3. Do not change sign convention halfway through a part.

BOTTOM LINE

A negative answer usually means opposite to your chosen direction.

Practice Page: SIGN

QUESTION TYPE Signs

A particle is projected upwards with speed $12\ \mathrm{m\,s^{-1}}$.

COMMON TRAP Treating deceleration as automatically negative without direction.

03 MEET: Meeting Motions

TRIGGER *two particles meet, overtake*

BECOMES Write each position from the same origin and set them equal.

FIRST LINE TO WRITE

$$s_A = s_B$$

SIMPLEST STRATEGY

1. **M**odel both positions.
2. **E**qualise positions.
3. **E**xtract valid time.
4. **T**ranslate time into the asked result.

WORKED MODEL

$s_A = 2t$, $s_B = 20 - t^2$, so meeting means $2t = 20 - t^2$.

HARD VARIANTS

1. Use a common origin and sign convention.
2. Reject negative times.
3. If they start at different times, adjust the time variable.

BOTTOM LINE

Meeting is same place at same time.

Practice Page: MEET

QUESTION TYPE Two-body motion

Two particles move on a line; find when one catches the other.

COMMON TRAP Equating speeds instead of positions.

04 VERT: Vertical Gravity

TRIGGER *thrown up, falling, maximum height*

BECOMES Use $a = -g$ if upward is positive and split at the top if needed.

FIRST LINE TO WRITE

$$v = 0 \quad \text{at greatest height}$$

SIMPLEST STRATEGY

1. **V**ertical positive direction.
2. **E**nter $a = -g$.
3. **R**ecognise top has $v = 0$.
4. **T**ime/height from SUVAT.

WORKED MODEL

At greatest height, $v = 0$, so $0 = u^2 - 2gh$.

HARD VARIANTS

1. Total flight may need two stages if launch and landing heights differ.
2. Use $g = 9.8$ only if numerical answers are wanted.
3. Velocity can be negative on the way down.

BOTTOM LINE

The top of the flight has zero velocity, not zero acceleration.

Practice Page: VERT

QUESTION TYPE Vertical motion

Find the greatest height of a particle projected upwards at 14 $\mathrm{m\,s^{-1}}$.

COMMON TRAP Setting $a = 0$ at the top.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
constant acceleration, u, v, a, s, t	$u,\ v,\ a,\ s,\ t$
opposite directions, deceleration	up/right/forward is positive
two particles meet, overtake	$s_A = s_B$
thrown up, falling, maximum height	$v = 0$ at greatest height

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
SUVAT: Choose The Missing Letter	Only use SUVAT when acceleration is constant. If direction changes, split the motion at $v = 0$. Convert units before writing the equation.
SIGN: Sign Convention	Displacement can be negative even when distance is positive. Final velocity sign tells direction. Do not change sign convention halfway through a part.
MEET: Meeting Motions	Use a common origin and sign convention. Reject negative times. If they start at different times, adjust the time variable.
VERT: Vertical Gravity	Total flight may need two stages if launch and landing heights differ. Use $g = 9.8$ only if numerical answers are wanted. Velocity can be negative on the way down.